

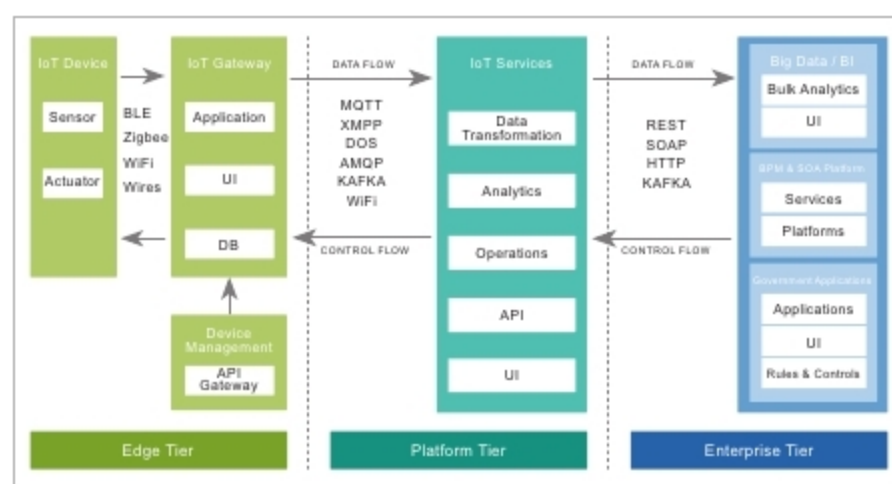


Figure 1: IoT reference architecture

Users of the IoT system include both humans and digital objects. Humans typically interact with the IoT system using smartphones, personal computers, tablets or specialised devices. In all cases, the IoT system provides some form of application that connects the human user with the rest of the IoT system.

The architecture allows:

- Connection, provision, and management of millions of distributed IoT devices and gateways with added security
- Simplification of data flow management with intelligence and analytics at the edge
- Provision of a comprehensive, centralised advanced analytics and data management platform with the ability to build or refine machine learning models and the pushing of these models to the edge
- To put security in the forefront, with pre-integrated security and manageability across devices, access, authentication and applications, as well as data that is in-motion and at-rest
- Utilisation, application development, deployment and integration services

Figure 2 depicts the open source IoT ecosystem.

There are many opportunities for enterprises to use IoT to make business services more efficient and customer-centric. IoT can help enterprises achieve their most important goals by:

- Significantly impacting the customer experience, and delivering a unified product and service experience
- Delivering huge cost savings
- Improving employee productivity
- Improving customer requests and service delivery
- Improving innovation

The open source IoT frameworks achieve efficient delivery and manage highly secure, reliable and scalable IoT solutions.

Without open standards and common protocols, the devices may not be able to speak to one another. While many IoT devices ultimately connect back to the Internet, the methods they use to communicate with one another and with local control hubs are often proprietary or poorly documented.

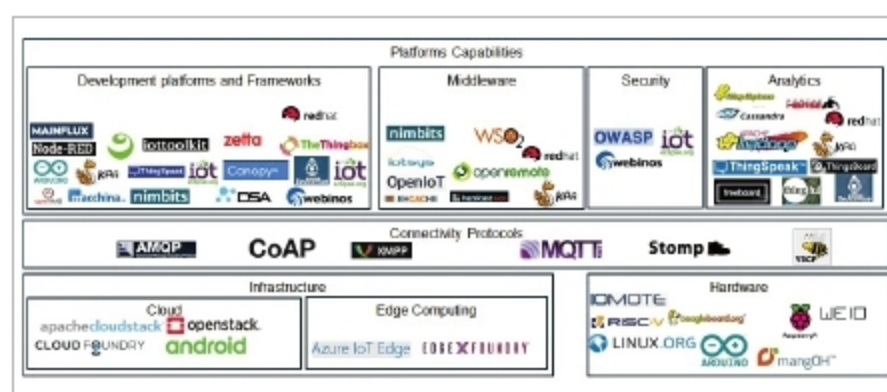


Figure 2: Open source IoT ecosystem

Open source technology enables permissionless, low-risk innovation that could accelerate the search for new cutting-edge use cases. For big enterprises and startups alike, open source represents the ability to explore new technologies without big, upfront financial commitments, and with fewer concerns over future royalties or intellectual property lawsuits.

Open source is an established reality in the Internet of Things. It is already being used by the vast majority of IoT developers — 90 per cent of them use open source technology at least occasionally, while 60 per cent also contribute to open source projects.

We can find high quality open source alternatives for any type of proprietary tool or technology used by developers. This can be an opportunity or a threat. It is a threat if you have a closed source offering that faces competition from an open source option that's also free.

Open source represents a giant opportunity for enterprises—a way to gain efficiencies and stimulate innovation by commodifying the non-core, non-differentiating parts of the technology stack, just like standards are. **END** 🐧

Acknowledgements

The authors would like to thank Hari Kishan Burle and Raju Alluri of the Global Enterprise Architecture Group of Wipro Technologies for their time and support to bring out this article as part of the Architecture Practice efforts.

References

- [1] "Gartner Says It's the Beginning of a New Era: The Digital Industrial Economy," Gartner, <http://www.gartner.com/newsroom/id/2602817>
- [2] <http://www.bharatexhibitions.com/en/OSI2018/spkrs/Col%20Inderjeet%20Singh.pdf>
- [3] https://www.w3.org/WoT/IG/wiki/images/9/95/The_essential_guide_to_Open_Source_in_IoT_-_White_Paper.pdf

By: Dr Gopala Krishna Behara and Raju Myadam

Dr Gopala Krishna Behara is a lead enterprise architect in the Global Enterprise Architecture division of Wipro. He can be reached at gopalkrishna.behara@wipro.com.

Raju Myadam is a chief enterprise architect in the Wipro Digital division of Wipro. He can be reached at raju.myadam@wipro.com.